

Preference-Aware Coefficient Correction for Rewarded-Soups-Style Model Merging



Motivation and Goal

$$\theta_i \rightarrow \delta_i = \theta_i - \theta_0 \rightarrow R \rightarrow \lambda^* = f(p, R)$$

Motivation:

User preferences p are defined over objectives, but model merging happens in task-vector space.

Goal:

Use task-vector geometry R to compute preference-aware merge coefficients λ^* .

Architecture Overview + Method

$$D = [\delta_1, \dots, \delta_m]$$

- Base model: **TinyLlama-1.1B**
- Five HelpSteer2 LoRA specialists
- Compute relationship matrix:

$$R = D^\top D$$

- Compute corrected coefficients:
- Merge adapters and evaluate with **ArmoRM**

$$\lambda^* = f(p, R)$$

main.tex Progress

- LoRA geometry section updated
- Notation cleanup: D , R , task vectors
- Reward-sensitivity / proxy theory added
- Added new method: C3

Preference-Alignment CAGrad-inspired Mapping

- Still working on reward guarantee / non-worsening theory

Implementation Progress

- TinyLlama HelpSteer2 workflow prepared
- Manual AdamW loop
- LoRA regularization:
 - $r=8$
 - $\alpha=16$
 - dropout 0.1
 - weight decay 0.01
- Warmup: 6%
- LR scheduler: first epoch normal, then decay
- Equalized data amount across attributes
- Export notebook for training graphs

Training Difficulties

- Helpfulness epoch $\approx$ 29 minutes
- Overfitting observed:
 - train loss keeps decreasing
 - eval loss increases later
- Response:
 - add LR decay
 - add Warmup

ArmoRM Monitoring

- Read-only diagnostic monitoring
- Not used as training loss
- Not used for checkpoint selection
- 20 prompts evaluated every 200 steps
- Output: reward vs global step graph

Computational Complexity

Operation	Cost	Note
Inner product $\delta_i^\top \delta_j$ (d -vectors)	$\mathcal{O}(d)$	d -dependent
Matrix product $D^\top D$ ($d \times m$ times $m \times d$)	$\mathcal{O}(m^2 d)$	inner dimension d
Matrix-vector $R\lambda$ ($m \times m$ times m)	$\mathcal{O}(m^2)$	d -free
Dense factorization / solve of an $m \times m$ system	$\mathcal{O}(m^3)$	Cholesky/LU
Projection onto the simplex Δ_m	$\mathcal{O}(m \log m)$	sort-based
Convex QP / QCQP over Δ_m (m vars), interior point	$\mathcal{O}(m^3)$ per iter	T_{ip} iters
Linear combination $\sum_i \lambda_i \delta_i$ (m d -vectors)	$\mathcal{O}(md)$	d -dependent

Computational Complexity

Method	Form	Per-preference cost (in m)
P1	closed form	$\mathcal{O}(m^2)$
M1	convex QP	$\mathcal{O}(m^3)$
M2	convex QP	$\mathcal{O}(m^3)$
P2	R -projections + QP	$\mathcal{O}(m^3)$
C2	smooth concave (first order)	$\mathcal{O}(m^2)$ per iter
C1	convex QCQP (interior point)	$\mathcal{O}(m^3)$ per iter
Precomputation of R (once)		$\mathcal{O}(m^2d)$
Realization $\theta(\lambda)$ (per chosen λ)		$\mathcal{O}(md)$

Next Steps

- Finish all 5 adapter trainings
- Save logs/adapters
- Generate train/eval/ArmoRM graphs
- Compute R
- Run proxy validation
- Evaluate coefficient mappings
- Update thesis text

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